

## 《The Imitation Game》Movie Reflection Report

By: 蔡定宇 (112006265)

I like *The Imitation Game*, but the part that I've been curious about is the cryptography:

- intercept the traffic
- guess a crib (simple, predictable words)
- turn that guess into wiring constraints
- run the Bombe to rule out impossible rotor settings
- test the survivors, then decrypt and decide what to do

I really enjoyed the movie and it actually gave me motivation to take Cryptography seriously, but like most cinematic portrayals some details aren't really realistic, for example:

The Bombe isn't a general computer. It's an electromechanical way to check implications fast. Welchman's diagonal board speeds that up. The film mentions none of this in detail, which is fine for pacing, but it's how the win actually happened.

Also, credit didn't start at Bletchley: the Polish team's earlier work also played a role in making a difference. And while the movie centers Turing, there were parallel efforts (Tutte, Flowers) on different ciphers. The way the movie portrayed was a single breakthrough but in fact it was teamwork and process that lead them to success. Although the movie is not completely accurate to history, it does leave inspiration of perseverance which I think is enough to leave an impact on viewers.

What the movie does get right is that "decrypt → decide" is the hard part. Reading messages is only useful if you act at the right time. Sometimes that means not using what you know to avoid exposing the source. That's uncomfortable, but it's part of real operations.

There was a portion in the movie where they drawn parallels between what humans imply during communication is in ways similar to encryption (hence the title "The Imitation Game", at least that's my take) that had me thinking about which particular aspects of everyday life I can notice more (etiquette). We speak with keys (context, tone, timing). Inside jokes only work if you share the key. "We're fine" after an argument is basically ciphertext; the crib is what happened earlier. Misreads are key mismatches. Apologies are key exchange. It's the same shape: use context to form a hypothesis, test it against the signals you have, then respond.

If I had to write my takeaway as a checklist for myself, it's this:

- define the pipeline
- name the cues (cribs) that lower the search space
- remember the Bombe is constraint pruning, not thinking
- decisions matter as much as decryption

The movie has mixed reviews but I believe being able to appreciate the good parts of anything is key.